Building a Chemical Language Model for RNA sequences with Masked Language Modelling

Rohan Ghosh Dastidar
Department of Chemical Engineering
Indian Institute of Technology Kharagpur, India
rgdastidar2069@kgpian.iitkgp.ac.in

***Abstract*—We pretrain a 4.9M-parameter transformer encoder on Rfam seed sequences with masked nucleotide prediction and evaluate its frozen embeddings with a convolutional base-pair probe trained on ViennaRNA partition-function probabilities. Evaluation uses a family-held-out split with cluster-level separation, and includes an architecture-matched random-initialization control, a canonical-pairing rule, a stem-pattern rule, and a shuffled-target control. On held-out sequences from seen families, the pretrained encoder reaches AUPRC 0.789 and AUROC 0.9965, compared with 0.777 and 0.9966 for the random control. On families unseen during pretraining, the values are 0.842 and 0.9973 for the pretrained encoder and 0.837 and 0.9972 for the control. Rule-based baselines reach AUPRC 0.015 to 0.035, and shuffled targets reduce the probe to near chance. Most of the probe's performance therefore comes from the convolutional head operating on sequence-derived features, and pretraining contributes a small gain of about 0.01 AUPRC that comes from a single seed.**

I. Introduction

RNA function heavily depends on it's secondary structure, where the base pairs fold a sequence into stems, loops, and bulges [1]. Computational methods such as the McCaskill partition function estimate these pairings from thermodynamic parameters [2], but they are expensive for large-scale analysis. Masked language models trained on biological sequences have been shown to learn residue-level information that correlates with structure in proteins [3] and, more recently, in RNA [4]. This work asks a narrower question - does a small transformer trained only to predict masked nucleotides produce embeddings from which base-pairing probabilities can be obtained, and does that information transfer to RNA families it did not see during training? We answer with controls designed to separate what pretraining adds from what the probe architecture contributes.

II. Methods

A. Data

We used the Rfam seed alignments, current release Rfam 15, which contain hand-curated family alignments with family accession labels [5]. Gap characters were removed, sequences were uppercased and mapped from T to U, and only sequences of 20 to 512 nucleotides were kept. Exact duplicates were also removed. After data cleaning, 97,407 sequences from 4,192 families remained.

B. Splitting the dataset

Near-duplicate homologs in Rfam seed alignments would inflate the probe performance if they appeared in both train and test sets. We therefore clustered sequences with MMseqs2 at 80% sequence identity and 80% coverage [6], and assigned each cluster as a unit. Approximately 10% of families (421 families, 8,690 sequences) were held out entirely as a novel-family test set. All other clusters were assigned to pretraining (60%), probe training (20%), or seen-family test (20%). Table I summarizes the splits.

TABLE I
Dataset splits, sizes, and architectural roles.

Split	Sequences	Families	Role
pretrain	53,896	3,080	MLM pretraining (2% held out for validation)
probe_train	17,326	1,786	Probe training (10% held out for validation)
probe_test	17,495	1,767	Seen-family test
novel_test	8,690	421	Unseen-family test

C. Model

The encoder is a BERT-style transformer with nucleotide tokens (A, C, G, U, N) along with [PAD], [CLS], and [MASK] special tokens. It uses learned token and absolute position embeddings, pre-LayerNorm blocks, 6 layers, model width 256, 8 attention heads, feed-forward width 1024, and a maximum input length of 513 tokens including [CLS]. The transformer is built with a total of 4,874,760 trainable parameters.

D. Pretraining

The model was trained with masked language modelling. 15% of nucleotide positions were selected; of those, 80% were replaced with [MASK], 10% with a random nucleotide, and 10% left unchanged [7]. We used AdamW (learning rate = 3×10^{-4}, weight decay = 0.01), batch size = 64, linear warmup over 5% of steps, and gradient clipping at 1.0. Training ran for 20 epochs. The validation loss fell from 1.30 to 1.18 and was still decreasing at the final epoch, so the model is undertrained relative to convergence.

E. Base-pair probe

Structure targets were computed with ViennaRNA's McCaskill partition function [2], giving per-pair probabilities. The encoder was frozen, and a bilinear pair head was trained on its final hidden states:

$$s_{ij} = \frac{(W_q h_i)^T (W_k h_j)}{\sqrt{d}} \tag{1}$$

symmetrized. Pairs with $|i - j| < 4$ were excluded. The loss was binary cross-entropy over valid pairs against the soft ViennaRNA probabilities. The probe was trained for 30 epochs with AdamW (learning rate 10^{-3}), and the checkpoint with the lowest validation loss was retained. Evaluation used the first 500 sequences of each test set, which yields 2.4 M and 1.9 M valid pairs for seen and novel families, respectively.

F. Baslines

Three references were used. (1) An architecture-matched encoder with random initialization, passed through the identical probe. (2) A canonical-pairing rule that scores a pair as 1 if the two bases form an AU, GC, or GU pair and 0 otherwise. (3) The base rate of positive pairs, which is the AUPRC of a random predictor. We report AUPRC, the metric most sensitive to the rare-positive regime here, with positives defined as probability above 0.5.

III. Results

TABLE II
Base-pair probe performance

Model	Seen AUPRC	Seen AUROC	Novel AUPRC	Novel AUROC
Base rate (chance)	0.006	—	0.008	—
Canonical-pairing rule	0.015	—	0.019	—
Stem rule	0.027	0.837	0.035	0.843
Random-init encoder + 2D probe	0.777	0.9966	0.837	0.9972
Pretrained encoder + 2D probe	0.789	0.9965	0.842	0.9973
Pretrained – random	+0.012	−0.0001	+0.005	+0.0001

With batched evaluation, which includes padding, both encoders score higher: random 0.801 and 0.857, pretrained 0.811 and 0.863 (seen and novel AUPRC). The gap between the two models is similar under both scoring schemes. The convolutional probe reaches AUPRC near 0.8 on both test sets, about 30 times the stem rule and 50 times the canonical rule. Pretraining adds about 0.01 AUPRC on seen families and 0.005 on novel families, with no change in AUROC. The shuffled-target control drops AUPRC to about 0.012 to 0.015, roughly twice the base rate, showing that the probe depends on the input sequence.

IV. Discussion

The main finding is that the probe architecture, not pretraining, accounts for nearly all of the measured performance. A convolutional head on random transformer features already reaches AUPRC 0.78 and 0.84, and pretraining adds only a small increment. The stem rule, which captures the most obvious pattern of stacked complementary pairs, scores 0.03. The convolutional head therefore learns a great deal beyond simple complementarity, but we cannot say from these experiments how much of that comes from the encoder's features rather than from the head learning sequence-to-structure patterns directly from its input.

The shuffled-target result shows that the probe is not producing generic scores. Its predictions are tied to the sequence and its thermodynamic structure. The residual signal, about twice the base rate, is expected because shuffled partners of the same length share composition and some stem-like patterns with the input.

The pretraining gain is small and comes from one seed. Its direction is consistent with a modest benefit, but we did not measure seed variance, so it should not be read as established. The gain on novel families is also small enough that it cannot be distinguished from noise on these data.

The labels are ViennaRNA predictions, not experimental structures, so the probe is reproducing a thermodynamic model. That is a legitimate target for representation evaluation, but it does not establish agreement with in-vivo or crystallographic structure.

V. Limitations

1) **Single seed.** Probe and pretraining seeds were not varied, so the pretraining gain has no measured uncertainty.
2) **Padding in batched evaluation.** The convolutional head sees padded positions in batched inputs, and this inflates scores slightly. All headline results use per-sequence scoring. The training loop was also batched, so the head was trained under the same padding condition.
3) **Evaluation subsets.** Each test set was evaluated on its first 500 sequences, not a random sample.
4) **Undertrained-to-converged encoder on a small corpus.** Pretraining used about 53,000 sequences from Rfam seeds and reached a plateau with a train-validation gap of about 0.1. Larger and more diverse corpora were not tested.
5) **Frozen encoder only.** No end-to-end fine-tuning was performed, so these results do not measure the upper bound for these architectures.
6) **Label source.** ViennaRNA probabilities are thermodynamic approximations, and positives are defined by a 0.5 threshold on those probabilities.
7) **No published-model comparison.** RNA-FM was not evaluated under the same protocol.

VI. Conclusion

A convolutional base-pair probe on frozen features from a small masked nucleotide language model reaches AUPRC near 0.8 on held-out families, far above rule-based references and sensitive to the input sequence. Most of that performance is attributable to the probe architecture: a randomly initialized encoder scores almost identically. Masked pretraining gives a small, single-seed improvement in AUPRC. Establishing whether pretraining contributes meaningfully will require multiple seeds, a larger pretraining corpus, and end-to-end fine-tuning on structure labels, which we identify as the main directions for future work.

References

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